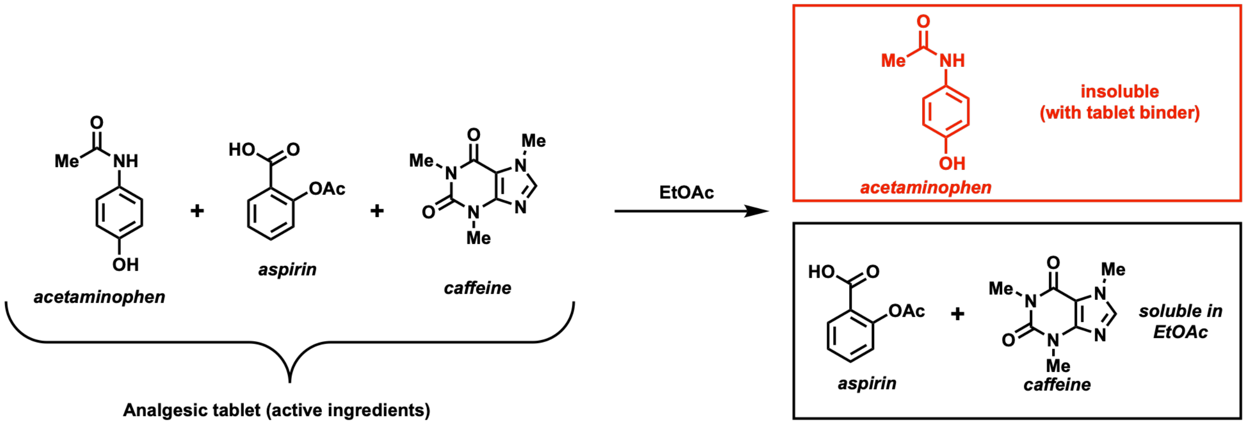
Activity 4: Analysis of Medicinally Active Organic Compounds in Over-the-Counter Pain Relievers: An Introduction to Chromatography

Figure 1. Pain Medication [1]

The Experiment

This laboratory exercise will highlight the utility of thin-layer chromatography to analyze compound mixtures by quickly and conveniently separating very small amounts of sample (microgram to sub-microgram). Specifically, we will use thin-layer chromatography techniques to provide a preliminary identification of the organic compounds (acetaminophen, aspirin, and caffeine) that are the medicinally active ingredients in Excedrin®, an over-the-counter pain reliever. To do this, we will compare the TLC behavior (R_f values) of the components of a pain reliever mixture with authentic samples of acetaminophen, aspirin, and caffeine. Moreover, we will examine how changing the polarity of the mobile phase effects the separation of the organic compounds in the mixture.

Activity 4: Procedure



Lab Activity Goal

Use thin-layer chromatography to analyze the medicinally active organic compounds present in an over-the-counter analgesic agent (pain relief medication).

Pre-Laboratory Assignments

See the Activity 4 Pre-Lab Preparation and Outline document in Labflow

Experimental Procedure

Part 1. Determine the Best TLC Solvent to Resolve the Analgesic Components of the Pain Reliever Tablet

Note: If the mortar and pestle is in use, then begin by preparing the TLC chambers of step 4.

1. Prepare a methanol (MeOH) extract of the over-the-counter pain reliever.
   a. Crush two tablets thoroughly using a mortar and pestle. Place the resulting powder into a 25-mL Erlenmeyer flask.
   b. Add ~5 mL of methanol and swirl for 5 minutes. Allow the undissolved solids to settle to the bottom of the flask (the undissolved material is tablet binding agents). This is the tablet extract that will be spotted

onto your TLC plates.

c. Labeling your MeOH extract: as: “T” for tablet is suggested.

2. Obtain a TLC plate that is large enough to accommodate 4 lanes. See Labflow video Running a TLC Analysis.

3. Spot the MeOH tablet solution (solution “T”) onto the first lane of three separate prepared TLC plates. Remember to follow your instructor’s directions for the spotting technique. Make sure to NOT spot the sample too heavily; too much analyte will streak the TLC plate during developing and make analysis impossible.

Pro-tip **Before developing a UV-active TLC plate, check the concentration of the baseline spots to ensure they are visible without being too concentrated.**

4. Prepare three separate mobile phase solvent systems (12 mL of each). Add each system to properly labeled 250 mL beakers (cover each beaker with a watch glass). (Note: Ethyl acetate is commonly abbreviated EtOAc).
   a. Use a graduated cylinder to prepare 12 mL of:
      (i) 5:1 hexanes:EtOAc
      (ii) 1:1 hexanes:EtOAc
      (iii)100% EtOAc

   b. Finish preparing the three separate TLC developing chambers. Be sure each chamber is labeled (e.g., “5:1”, “1:1” & “0:1”). To each mobile phase solvent system, your instructor will add glacial acetic acid (5 drops) and swirl the beaker. The addition of acid prevents the aspirin from streaking on the TLC plate.

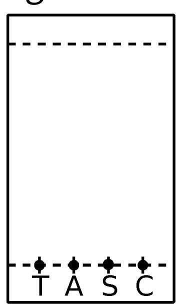
Activity 4: Procedure

Important: Safety Pause

Capillary tubes are made of glass and can easily break and cause cuts if you are not cautious. Store any capillary tubes not in use upright in a 25 mL Erlenmeyer flask to prevent accidental breakage!

Safety Pause

5. With a new capillary tube, spot the three remaining labeled lanes on each of your plates with the analgesic standards (already prepared). It is important not to contaminate these standards by using used capillary tubes or mixing up the caps on the standards. These standards will be acetaminophen (A), aspirin (S), and caffeine (C). Each plate should look like the following diagram:



T = Tablet
A = Acetaminophen
S = Aspirin
C = Caffeine

6. Develop the three identical TLC plates in each of the three different developing solvents. Be sure to IMMEDIATELY trace the solvent front on the developed TLC plate with a pencil after removal from the TLC chamber.

7. Visualize the developed plates using UV lamps (as directed by your instructor).
   - Circle the spots that become visible under the UV light.
   - Measure and record the R_f of each spot on your plates.

Activity 4: Procedure

Important: Safety Pause

NEVER look into the UV light. Doing so can cause damage to your eyes!

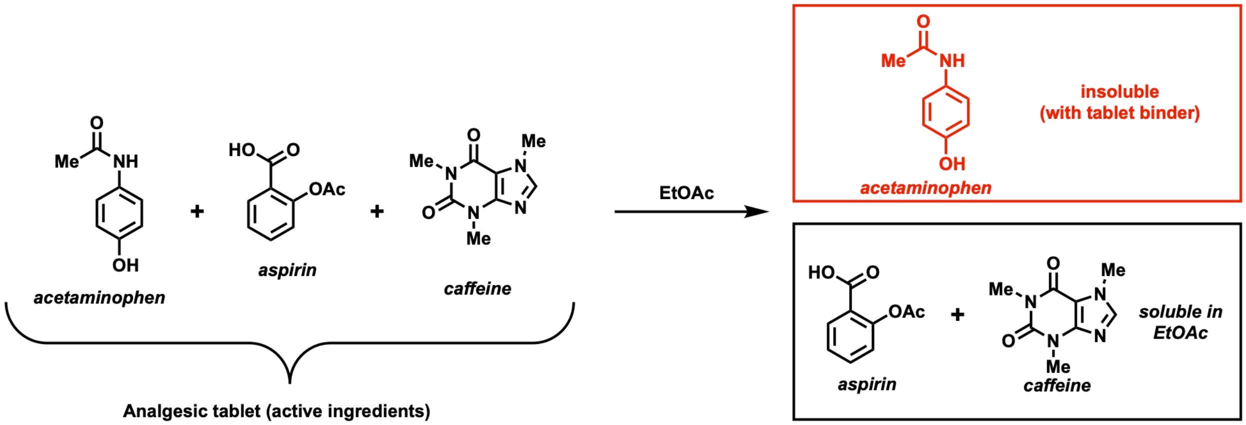
Safety Pause

8. Carefully record your observations of the developed TLC plate noting how the R_f values of the authentic samples (acetaminophen, aspirin & caffeine) compare to the R_f values of the pain reliever tablet extract (T).

9. Does this experiment provide preliminary evidence that the three components of the pain reliever tablet are acetaminophen, aspirin, and caffeine? If so, proceed to Part 2 of the experiment.
   - DO NOT throw away the “T” tablet MeOH solution; you will use it in Part 2.
   - DO NOT throw away the TLC solvent that afforded the best separation of the three analgesics in the tablet; you will use it in Part 2.

Activity 4: Procedure

Part 2. Separate & Isolate Acetaminophen (A) from a Pain Reliever Tablet

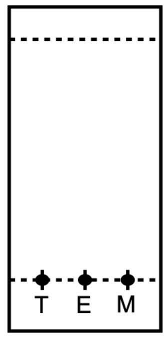


Acetaminophen, unlike aspirin and caffeine, is virtually insoluble in ethyl acetate (CH_3CO_2Et), also referred to as EtOAc. Thus, acetaminophen can be separated from aspirin and caffeine via a solid-liquid extraction. Accordingly, when a crushed pain reliever tablet is swirled in EtOAc, acetaminophen should remain undissolved (along with the tablet binder) while the aspirin and caffeine readily dissolve in the EtOAc solvent. Let's test the validity of this separation strategy using TLC to analyze our success.

1. Place one crushed pain reliever tablet into a 25-mL Erlenmeyer flask.
   a. From your instructor's specified location, carefully obtain 15-20 mL of ethyl acetate from a stock beaker (which has been prepared by your instructor).
   b. Add ~10 mL of EtOAc to the flask containing your crushed tablet and swirl for 10 min. The remaining ethyl acetate will be used in step (d) below.
   c. Use a funnel, a 25-50 mL Erlenmeyer, and fluted filter paper to gravity filter the EtOAc tablet mixture.
   d. Rinse the solid trapped in the filter paper with another ~5 mL of EtOAc.
   e. Reserve the EtOAc filtrate for later TLC analysis. Be sure to label it "E."
2. Remove the filter paper containing the solid material from the funnel and allow it to air dry. Dissolve a small portion of this solid in a 50-mL beaker using ~ 2 mL of MeOH. Reserve (and label "M") this solution for TLC analysis.

Activity 4: Procedure

3. Prepare a TLC plate for the analysis by tracing an origin line and marking 3 lanes. Your plate should look something like the following diagram:



T = Tablet Solution
E = EtOAc Extract
M = Methanol Solution

Label each lane appropriately:
(i) The first lane should be the tablet solution (T) from Part 1.
(ii) The second lane should be the EtOAc extract (E) of a tablet.
(iii) Finally, the third lane should be MeOH solution (M).

4. Develop the TLC plate using the mobile phase solvent system affording the best separation of the three compounds from Part 1.

5. After developing the plate (and tracing the solvent front) visualize the plate (and circle all the spots).

6. Measure and record all the R_f values in your notebook.

7. Assess the effectiveness of this solid-liquid extraction technique for separating acetaminophen from the aspirin/caffeine mixture.

Part 3. Melting Point (and Dry Mass) Determination of Crude and Purified Benzoic Acid from Activity 3

1. As part of Activity 3, you isolated benzoic acid from a three-component mixture using extraction techniques. This isolated "crude benzoic acid" was then purified by recrystallization (from water) giving you "purified benzoic acid". During Activity 4, you will need to measure the melting point of each sample (crude and pure) and record the dry mass of purified benzoic acid.

2. Before you leave lab, upload your in-lab notebook pages to Labflow. Your in-lab notebook pages must include the following in a **single pdf file**:
   - Any experimental observations (including but not limited to: color changes, observational notes, measurements, sketches/images of your TLC plates, etc.)
   - Your handwritten answers to the questions posted in the In-Lab Notebook Pages tile on Labflow
   - A picture of your Organic Lab Cleaning Checklist

Clean-Up and Waste Disposal

Activity 4 Specific Cleanup

- **Organic Liquid Waste**
  - Acetone rinses
    - Rinse your glassware with a <u>small</u> amount of wash acetone (squeeze bottles with red caps) and dispose of acetone in the liquid waste
  - TLC mobile phases (ethyl acetate and hexane)
  - Methanol
  - Ethyl acetate (EtOAc)
  - Dissolved tablet solution (T)
  - EtOAc filtrate (E)
  - Dissolved methanol solution (M)

- **Solid Waste (Red Can)**
  - Used gloves
  - Used TLC plates
  - Used filter paper
  - Leftover crushed tablets (if any)
  - Remaining crude and/or purified benzoic acid from Activity 3
- **Glass Waste**
  - Melting point tubes
  - Capillary tubes
    - Check the foil (the metal lip in the front part of your hood) for runaway capillary tubes and dispose of them if present.

Routine Cleanup

- Clean up your hood space of all trash, spilled chemicals, etc. Don't leave a mess for the people who share this hood at other time slots. Up to 24 other people are sharing this space at various time slots throughout the week. Clean up your mess!
- No dirty or clean glassware should be left in the sink. No used filter paper or wadded up dirty paper towels should be left in the sink.
- Dirty paper towels go into the REGULAR TRASH not the red solid waste trash. This saves the university a lot of waste disposal money!
- Dispose of dirty filter paper in the red solid waste container.
  - Do NOT deposit filter paper in the liquid waste eco-funnel as this will cause a clog which will lead to a messy overflow of waste.
  - Do NOT leave filter paper in the sink as this is poor lab etiquette and rude.
  - Used gloves should be disposed of into the red solid waste.
- Ask your instructor if you are in charge of cleaning up the balance area for this week. If you are in charge do the following:
  - Be sure all reagent bottles are properly capped.
  - Use the small brush to clean off balances.
  - Use small broom and dustpan (located on the wall surrounding the balance area) to remove excess solid from counter. Dispose of excess solids in red solid waste container. Return broom and

    dustpan to proper location.
    - Wipe down the counter with a moist paper towel.
- Ask your instructor if you are in charge of making sure the sink area is presentable for the next lab group.
    - Wash any dirty glassware rudely left in the sink. Let clean glassware air dry on rack or windowsill area.
    - Dispose of any paper towels, filter paper, or cotton rudely left in the sink.
    - Do not remove broken glass from the drain or the sink. Rather, tell your instructor and then fill out a "something wrong form" located in the cabinet above the solid waste containers. Be sure to record room number and the Left/Right location of the sink.

Post-Lab Assignment

The Activity 3 Post-Lab Assessment ***and*** Activity 4 Post-Lab Assessment are due at the beginning of next week’s lab.

Image Credits

[1] Image courtesy of Pixabay under CC0 1.0